

## SAIDA, Algeria

WMO: 605360

Lat: 34.8913N

Lon: 0.1577E

Elev: 752

StdP: 92.61

Time Zone: 1.00 (E01)

Period: 01-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.6       | 0.7 | -4.8                          | 2.8 | 9.2  | -3.1 | 3.2 | 7.8  | 15.7                  | 8.6  | 13.5 | 9.0  | 0.4                   | 0    | 0.471 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.3                   | 39.2            | 19.9 | 37.7 | 19.7 | 36.2 | 19.5 | 22.6                | 33.1 | 21.7 | 32.5 | 20.9 | 31.8 | 5.4                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.6                            | 15.7 | 26.6 | 18.6 | 14.7 | 26.1 | 17.6 | 13.8 | 25.6 | 70.6          | 32.9 | 66.9 | 32.4 | 64.1 | 31.8 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 12.5              | 10.4 | 8.9 | -3.4                       | 41.6 | 2.8                | 0.8 | -5.4                                               | 42.2 | -7.1       | 42.6 | -8.6       | 43.1 | -10.7      | 43.7 |    |    |
|                   |      |     | -4.0                       | 24.2 | 2.7                | 1.2 | -6.0                                               | 25.1 | -7.6       | 25.8 | -9.1       | 26.5 | -11.1      | 27.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 17.2   | 8.7   | 9.0   | 11.9  | 14.5  | 18.5  | 23.8  | 28.1  | 27.5  | 23.0  | 19.3  | 12.7  | 9.3  |
|                                                                   | DBStd       | 7.68   | 2.83  | 3.37  | 3.50  | 3.91  | 4.27  | 3.96  | 2.84  | 2.59  | 3.22  | 3.58  | 3.56  | 3.22 |
|                                                                   | HDD10.0     | 201    | 60    | 52    | 21    | 6     | 0     | 0     | 0     | 0     | 0     | 0     | 12    | 49   |
|                                                                   | HDD18.3     | 1428   | 299   | 261   | 201   | 127   | 52    | 3     | 0     | 0     | 2     | 30    | 173   | 280  |
|                                                                   | CDD10.0     | 2846   | 19    | 24    | 79    | 140   | 265   | 415   | 561   | 543   | 389   | 289   | 93    | 28   |
|                                                                   | CDD18.3     | 1032   | 0     | 0     | 1     | 12    | 58    | 168   | 302   | 284   | 141   | 61    | 4     | 1    |
|                                                                   | CDH23.3     | 13415  | 0     | 3     | 32    | 222   | 808   | 2235  | 4168  | 3766  | 1516  | 634   | 28    | 2    |
| Wind                                                              | WSAvg       | 2.8    | 2.7   | 3.1   | 3.1   | 3.0   | 2.8   | 2.7   | 2.7   | 2.6   | 2.4   | 2.5   | 2.9   | 2.7  |
|                                                                   | PrecAvg     | 352    | 39    | 38    | 38    | 41    | 32    | 12    | 6     | 10    | 20    | 37    | 49    | 31   |
|                                                                   | PrecMax     | 531    | 104   | 124   | 172   | 124   | 100   | 70    | 31    | 45    | 79    | 146   | 188   | 81   |
|                                                                   | PrecMin     | 171    | 0     | 0     | 0     | 2     | 3     | 1     | 0     | 1     | 0     | 0     | 12    | 0    |
|                                                                   | PrecStd     | 89     | 31    | 26    | 37    | 29    | 22    | 16    | 7     | 11    | 18    | 34    | 34    | 23   |
|                                                                   | DB          | 21.3   | 22.8  | 26.7  | 30.7  | 35.4  | 39.2  | 41.2  | 40.5  | 36.6  | 33.6  | 26.7  | 22.4  |      |
|                                                                   | MCWB        | 11.0   | 12.2  | 13.8  | 15.1  | 18.2  | 19.5  | 20.1  | 19.7  | 19.0  | 18.1  | 15.4  | 12.8  |      |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 18.5   | 19.5  | 23.5  | 28.2  | 32.6  | 37.2  | 39.6  | 38.8  | 34.5  | 31.3  | 23.4  | 18.7  |      |
|                                                                   | 2% MCWB     | 10.1   | 11.1  | 13.3  | 14.8  | 17.5  | 19.2  | 19.9  | 20.3  | 19.0  | 17.3  | 13.5  | 11.4  |      |
|                                                                   | 5% DB       | 16.1   | 17.3  | 21.1  | 25.9  | 30.3  | 35.3  | 38.1  | 37.4  | 32.7  | 29.2  | 21.1  | 16.7  |      |
|                                                                   | 5% MCWB     | 9.3    | 10.2  | 12.8  | 14.7  | 17.0  | 18.8  | 19.8  | 19.9  | 19.0  | 16.8  | 13.1  | 10.6  |      |
|                                                                   | 10% DB      | 14.0   | 15.2  | 19.2  | 23.3  | 28.1  | 33.3  | 36.4  | 35.8  | 30.8  | 26.9  | 19.0  | 14.7  |      |
|                                                                   | 10% MCWB    | 9.2    | 9.5   | 12.1  | 13.9  | 16.3  | 18.5  | 19.5  | 19.9  | 18.4  | 16.2  | 12.3  | 9.7   |      |
|                                                                   | WB          | 12.1   | 13.7  | 16.0  | 18.5  | 20.7  | 21.8  | 23.3  | 23.9  | 23.1  | 21.6  | 17.6  | 13.8  |      |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% MCDB   | 18.0   | 20.2  | 21.8  | 25.8  | 31.6  | 34.8  | 34.5  | 34.4  | 31.8  | 29.3  | 23.8  | 19.7  |      |
|                                                                   | 2% WB       | 11.2   | 12.1  | 14.6  | 16.4  | 19.0  | 20.6  | 22.2  | 22.9  | 21.4  | 18.9  | 15.5  | 12.3  |      |
|                                                                   | 2% MCDB     | 16.0   | 17.4  | 21.1  | 24.6  | 28.7  | 32.9  | 33.5  | 33.2  | 29.2  | 27.2  | 20.5  | 17.8  |      |
|                                                                   | 5% WB       | 10.2   | 10.9  | 13.6  | 15.3  | 18.0  | 19.7  | 21.3  | 21.9  | 20.4  | 17.8  | 14.0  | 11.1  |      |
|                                                                   | 5% MCDB     | 14.3   | 15.8  | 19.7  | 23.8  | 27.5  | 32.0  | 33.2  | 32.7  | 28.4  | 25.8  | 19.3  | 15.7  |      |
|                                                                   | 10% WB      | 9.4    | 9.9   | 12.5  | 14.3  | 17.0  | 19.0  | 20.6  | 21.1  | 19.6  | 17.0  | 12.6  | 10.2  |      |
|                                                                   | 10% MCDB    | 13.3   | 14.3  | 18.1  | 21.9  | 26.3  | 31.5  | 33.0  | 31.6  | 27.9  | 24.5  | 17.8  | 13.9  |      |
| Mean Daily Temperature Range                                      | MDBR        | 10.1   | 10.1  | 12.0  | 13.4  | 14.1  | 15.6  | 16.3  | 16.0  | 13.9  | 12.7  | 9.9   | 9.3   |      |
|                                                                   | 5% DB       | 14.1   | 14.2  | 16.0  | 18.4  | 17.8  | 17.6  | 17.2  | 17.5  | 16.3  | 15.5  | 13.1  | 12.3  |      |
|                                                                   | MCWBR       | 7.7    | 7.3   | 7.9   | 8.1   | 6.9   | 5.9   | 5.5   | 5.9   | 6.0   | 6.1   | 6.1   | 6.8   |      |
|                                                                   | 5% WB       | 11.7   | 12.2  | 14.1  | 16.5  | 15.6  | 16.1  | 15.6  | 16.1  | 14.4  | 13.4  | 11.5  | 11.0  |      |
|                                                                   | MCWBR       | 7.0    | 6.8   | 7.6   | 8.1   | 6.6   | 6.3   | 6.1   | 6.7   | 6.3   | 6.1   | 5.8   | 6.5   |      |
| Clear-Sky Solar Irradiance                                        | taub        | 0.292  | 0.322 | 0.388 | 0.412 | 0.450 | 0.476 | 0.536 | 0.499 | 0.456 | 0.395 | 0.340 | 0.303 |      |
|                                                                   | taud        | 2.495  | 2.409 | 2.191 | 2.144 | 2.061 | 1.989 | 1.835 | 1.931 | 2.052 | 2.218 | 2.385 | 2.487 |      |
|                                                                   | Ebn at Noon | 921    | 923   | 879   | 874   | 841   | 816   | 765   | 785   | 800   | 823   | 847   | 883   |      |
|                                                                   | Edh at Noon | 83     | 101   | 138   | 152   | 167   | 179   | 208   | 185   | 156   | 121   | 92    | 79    |      |
| All-Sky Solar Radiation                                           | RadAvg      | 2.88   | 3.75  | 5.02  | 6.07  | 6.90  | 7.71  | 7.59  | 6.89  | 5.53  | 4.33  | 3.06  | 2.58  |      |
|                                                                   | RadStd      | 0.26   | 0.36  | 0.28  | 0.45  | 0.45  | 0.25  | 0.16  | 0.25  | 0.31  | 0.28  | 0.24  | 0.20  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | +0.29 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page